Chapter 47

The Role of Sonography in Obstetrics

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Introduction

- Sonography is primary tool for evaluating developing fetus during pregnancy.
- Obstetric sonography allows clinician to assess fetal development, growth, and well-being.
- When abnormal condition recognized prenatally, obstetric management may be altered to provide optimal care for fetus and child.

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Introduction (Cont.)

- Prenatal diagnosis has led to prenatal treatments performed under ultrasound visualization.
- Sharing sonographic images and diagnostic results facilitates prenatal parental education and counseling.

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Introduction (Cont.)

- Sonographers performing fetal studies must understand both sonographic and obstetric principles.
- Sonographers must accurately and thoroughly compile pertinent information to provide optimal sonographic assessment of fetus.

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Guidelines

- Practice guidelines produced by American College of Radiology (ACR), American Institute of Ultrasound in Medicine (AIUM), and American College of Obstetricians and Gynecologists (ACOG) recommend specific components of a standard obstetric sonography examination.

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Indications for Obstetric Sonography

1. Estimation of gestational (menstrual) age for patients with uncertain clinical dates or verification of dates for patients who are to undergo scheduled elective repeat cesarean delivery, indicated induction of labor, or elective termination of pregnancy. Sonographic confirmation of dating permits proper timing of cesarean delivery or labor induction to avoid premature delivery.

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Indications for Obstetric Sonography (Cont.)

2. Evaluation of fetal growth, for example, when patient has identified cause for uteroplacental insufficiency, such as severe preeclampsia, chronic hypertension, chronic renal disease, or severe diabetes mellitus, or for other medical complications of pregnancy in which fetal malnutrition (e.g., intrauterine growth restriction [IUGR] or macrosomia) is suspected. Measuring fetal growth by sonography at 2- to 4-week intervals permits assessment of impact of complicating condition on fetus and guides pregnancy management.

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Indications for Obstetric Sonography (Cont.)

3. Vaginal bleeding of undetermined cause in pregnancy. Sonography often allows determination of source of bleeding and status of fetus.
4. Serial evaluation of cervical length in pregnant women with increased risk for recurrent or primary preterm birth

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Indications for Obstetric Sonography (Cont.)

5. Evaluation of abdominal or pelvic pain in pregnancy that may be associated with ectopic pregnancy, abruptio placenta, or maternal appendicitis, renal calculi, pelvic mass, or other conditions.

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Indications for Obstetric Sonography (Cont.)

6. Determination of fetal presentation when presenting part cannot be adequately determined in labor or fetal presentation is variable in late pregnancy. Accurate knowledge of presentation guides management of delivery.

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Indications for Obstetric Sonography (Cont.)

7. Suspected multiple gestation based on detection of more than one fetal heart beat pattern, fundal height larger than expected for dates, or prior use of fertility drugs. Pregnancy management may be altered in multiple gestation.

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Indications for Obstetric Sonography (Cont.)

8. Adjunct to amniocentesis, chorionic venous sampling (CVS), and other invasive procedures. Sonography permits guidance to the intended target.

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Indications for Obstetric Sonography (Cont.)

9. Significant discrepancy between uterine size and clinical dates. Sonography permits accurate dating and detection of such conditions as oligohydramnios and polyhydramnios, along with multiple gestation, IUGR, and anomalies.

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Indications for Obstetric Sonography (Cont.)

10. Evaluation of pelvic mass. Sonography can detect location and nature of mass and aid in diagnosis

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Indications for Obstetric Sonography (Cont.)

11. Hydatidiform mole suspected on basis of clinical signs of hypertension, proteinuria; presence of ovarian cysts felt on pelvic examination; or failure to detect fetal heart tones with a Doppler ultrasound device after 12 weeks. Sonography permits accurate diagnosis and differentiation of neoplasm from fetal death.

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Indications for Obstetric Sonography (Cont.)

12. Adjunct to cervical cerclage placement. Sonography aids in timing and proper placement of the cerclage for patients with incompetent cervix.
13. Suspected ectopic pregnancy or when pregnancy occurs after tuboplasty or prior ectopic gestation. Sonography valuable diagnostic aid for this complication.

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Indications for Obstetric Sonography (Cont.)

14. Evaluation of suspected fetal death. Rapid diagnosis enhances optimal management.
15. Suspected uterine abnormality (e.g., clinically significant leiomyomata, congenital structural abnormalities, such as bicornuate uterus or uteri didelphys). Serial surveillance of fetal growth and state enhances fetal outcome.

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Indications for Obstetric Sonography (Cont.)

16. Evaluation for fetal well-being. Biophysical evaluation for fetal well-being after 28 weeks of gestation may include assessment of amniotic fluid, fetal tone, body movements, breathing movements, and heart rate patterns.

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Indications for Obstetric Sonography (Cont.)

17. Evaluation of suspected amniotic fluid abnormalities such as suspected polyhydramnios or oligohydramnios. Confirmation of diagnosis and identification of cause of condition in certain pregnancies necessary

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Indications for Obstetric Sonography (Cont.)

18. Suspected abruptio placentae. Confirmation of diagnosis and extent of abruption assists in clinical management.
19. Adjunct to external version from breech to vertex presentation. Visualization provided by sonography facilitates performance of procedure.

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Indications for Obstetric Sonography (Cont.)

20. Estimation of fetal weight and presentation in premature rupture of membranes or premature labor. Information provided by sonography guides management decisions on timing and method of delivery.

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Indications for Obstetric Sonography (Cont.)

21. Evaluation following maternal serum biochemical marker results. Elevated alpha feto-protein (AFP) increases risk for open defects like neural tube defects. Other biochemical markers in first trimester or quad screen biochemistry in second trimester may indicate increased risk for certain obstetric or fetal conditions.

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Indications for Obstetric Sonography (Cont.)

22. Follow-up observation of identified fetal anomaly. Sonographic assessment of progression or lack of change may assist in clinical management.
23. Follow-up evaluation of placenta location for suspected placenta previa

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Indications for Obstetric Sonography (Cont.)

24. Evaluation for those with history of previous congenital anomaly. Detection of recurrence may be facilitated, or psychological benefit to patients may result from reassurance of no recurrence.

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Indications for Obstetric Sonography (Cont.)

25. Evaluation of fetal condition in late registrants for prenatal care. Assessment of gestational age and fetal size assists in pregnancy management decisions.

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Indications for Obstetric Sonography (Cont.)

26. To assess findings that may increase risk of aneuploidy
27. To screen for fetal anomalies through measurement of nuchal translucency or visualization of fetal structural anomalies

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Types of Obstetric Sonography Examinations

- Three practice guidelines define major types of sonographic examinations performed in second and third trimester of pregnancy using terms “limited,” “standard,” and “specialized” or detailed.

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Types of Obstetric Sonography Examinations (Cont.)

- 18 weeks
- Standard examination includes evaluation of gestational age by fetal biometry, fetal number, fetal presentation, placental position, cardiac activity, amniotic fluid volume, and fetal anatomic survey including all of elements specified in guidelines.

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Standard Exam (Cont.)

- May include maternal cervix and adnexa as well when clinically appropriate and technically feasible.
- If the cervix cannot be visualized, a transvaginal scan may be considered when evaluation or measurement of the cervix is needed.

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Types of Obstetric Sonography Examinations (Cont.)

- Limited obstetric sonography examination (CPT code 76815) used for answer to specific clinical question such as presentation of fetus, placental location, cervical length, amniotic fluid volume, or when verification of fetal heart motion required
- Is done when there is previous standard obstetric examination recorded

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Types of Obstetric Sonography Examinations (Cont.)

- Repeat obstetric sonography examination (CPT code 76816) similar to standard obstetric examination and typically includes biometry to evaluate fetal growth and reevaluation of anatomy that may or may not have been well visualized on standard examination
- Repeat examination done when there is previous standard obstetric examination recorded and second examination ordered for same indication

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Types of Obstetric Sonography Examinations (Cont.)

- Specialty obstetric sonography examination (CPT code 76811); also known as targeted sonogram
- Is typically done when anomaly suspected based on maternal history, biochemistry, or results of previous obstetric sonogram when there is a known fetal growth disorder, or when there is increased risk for a fetal condition

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Types of Obstetric Sonography Examinations (Cont.)

- Specialty obstetric sonography examination includes all components of standard examination plus more in-depth view of fetal anatomy.
- Includes additional views of fetal heart; may include color Doppler views of heart
- May also include additional views of extremities and focus on areas of anomalous or expected findings associated with patient history

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Types of Obstetric Sonography Examinations (Cont.)

- First trimester examination (CPT code 76801) performed before 13 weeks and 6 days of gestation
- Includes uterus, cervix, maternal adnexa as well as gestational sac and embryo
- Pregnancy dated based on embryonic size and fetal heart motion documented if these findings present

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First Trimester Exam (Cont.)

- Uterine anomalies and pelvic masses associated with pregnancy more easily seen in first trimester examinations
- Chorionicity and amnionicity of multiple gestations should be documented at this time as well.
- Exam of fetal anatomy should be performed.

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Types of Obstetric Sonography Examinations (Cont.)

- First trimester risk assessment examination (CPT code 76813); also known as nuchal translucency examination
- Is performed only when women choose first trimester screening tests for aneuploidy
- Includes measurement of fetal crown rump length and measurement of nuchal translucency using standard criteria

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Patient History

- Open-ended questions such as “Do you have concerns?” and closed questions such as “When was your last normal period?” are used in gathering important patient information.

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Gravidity and Parity

- Gravidity (G) is number of pregnancies, including present one
- Parity (P) reported using numeric system that describes all possible pregnancy outcomes
- Letter “P” followed by four numbers in sequence, P0000, commonly used

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Gravidity and Parity (Cont.)

- Numbers represent, in order, full-term deliveries, premature births and stillborns, early pregnancy loss or termination, and living children
- For instance, G4P2103 describes patient undergoing her fourth pregnancy. She has had two full-term deliveries, one premature birth, no early pregnancy losses, and three living children.

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Clinical Dates

- First date of last normal menstrual period (LMP or LNMP) is standard way to date pregnancy in United States.
- Human pregnancy lasts 266 days plus or minus 10 days.
- If conception occurs on day 14 from LNMP, pregnancy duration from LNMP is 280 days or 40 weeks.

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Clinical Dates (Cont.)

- Pregnancy divided into trimesters of approximately 13 weeks each
- Pregnant woman in first trimester until 13 weeks 6 days' gestational age (GA)
- Second trimester between 14 weeks and 26 weeks 6 days' GA
- Third trimester begins at 27 weeks' GA and lasts until term.

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Clinical Dates (Cont.)

- In reality, assessment of gestational age often not precise
- Many women have irregular periods, conceive within 3 months of coming off birth control pills when ovulation irregular, or do not record dates.
- Even with known menstrual date, conception may occur from day 6 to day 27, which is difference of 3 weeks in gestational age as determined by sonography.

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Clinical Dates (Cont.)

- Sonographer first asks patient first day of her last menstrual period.
- If patient does not remember date of her last normal menstrual period, may ask for expected date of delivery (EDD)
- Should also ask if there have been previous sonographic examinations performed before 20 weeks and estimated date of delivery determined by earliest sonographic examination

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Clinical Dates (Cont.)

- The sonographer should not change the EDD or “redate” the pregnancy once an earlier sonogram has established gestational age.
- Dates established by sonography performed in first or second trimester typically take precedence over menstrual dates when discrepancy >7 days in first trimester or >10 days in second trimester.

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Clinical Dates (Cont.)

- Sonography may be considered to confirm menstrual dates if gestational age agreement within week by crown-rump length or 10 days by second trimester fetal biometry.
- Pregnancy should not be dated by sonographic measurements in third trimester.
- Dates should not be changed after having been calculated from early exam.

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Nägele's Rule

- EDD = LMP - 3 months + 7 days
- LMP = EDC + 3 months - 7 days
- For example, LMP of 10/17 would result in EDD of 7/24 (10/17 - 3 months = 7/17 + 7 days = 7/24)

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Maternal Risk Factors

- Ask if patient has latex allergies.
- Ask if patient has experienced supine hypotension, i.e., difficulty lying on back during pregnancy.
- Caution patient to report any sense of warmth, dizziness, or syncope during sonography examination.

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Maternal Risk Factors (Cont.)

- Sonographer needs to know if patient currently taking any medication or has experienced clinical problems with pregnancy, such as:
  - Bleeding
  - Decreased fetal movement
  - Pelvic pain

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Maternal Risk Factors (Cont.)

- If patient has had problems with previous pregnancies, such as incompetent cervix, fibroids, fetal macrosomia or growth restriction, or congenital or chromosomal fetal anomalies, this information must be documented.

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Maternal Risk Factors (Cont.)

- Sonographer needs to know if patient has maternal risk factors for anomalies that may impact examination or interpretation.

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Maternal Risk Factors (Cont.)

- Factors that may affect risk of producing fetus with congenital anomalies include increased maternal age, first or second trimester maternal serum biochemistry values, first trimester increased nuchal translucency, maternal disease (e.g., diabetes mellitus and systemic lupus erythematosus), and pregnant uterine cavity either too small or too large for dates.

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Maternal Risk Factors (Cont.)

- Other risk factors include previous child born with chromosomal disorder or exposure to known teratogenic drug or infectious agent known to cause birth defects.
- Some anomalies caused by reduced or increased number of fetal chromosomes.
- These anomalies, most common of which is Down syndrome, trisomy 21, called aneuploidies

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Genomic Technology

- Genomic technologies have created multiple options for prenatal testing to screen and diagnose fetal anomalies.

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Safety of Ultrasound

- Animal experiments and therapeutic applications document that high-intensity ultrasound may modify biologic structures and functions.
- Studies in pregnancy with animals and humans have suggested possibilities of growth differences (reduction in animals), increased non-right handedness, and delayed speech.

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Safety of Ultrasound (Cont.)

- Major biologic effects of ultrasound
  - Thermal (rise in temperature)
  - Mechanical forces including cavitation (production and collapse of gas-filled bubbles)

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Safety of Ultrasound (Cont.)

- Sonographers can minimize thermal effects by not staying in one spot, especially over fetal bone, for long periods of time and by extending focus of beam as deep into body as reasonable to obtain adequate images.
- Cavitation dependent on presence of gas preexisting within tissue and is not an issue in the fetus.

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AIUM: *Prudent Use and Clinical Safety* Adopted in 2012

- Diagnostic ultrasound has been in use since late 1950s.
- Given its known benefits and recognized efficacy for medical diagnosis, including use during human pregnancy, AIUM herein addresses clinical safety of such use: *No independently confirmed adverse effects caused by exposure from present diagnostic ultrasound instruments have been reported in human patients in the absence of contrast agents.*

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AIUM: *Prudent Use and Clinical Safety* Adopted in 2012 (Cont.)

- Biological effects (such as localized pulmonary bleeding) have been reported in mammalian systems at diagnostically relevant exposures but clinical significance of such effects is not known.
- Ultrasound should be used by qualified health professionals to provide medical benefit to patient.

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AIUM: *Prudent Use and Clinical Safety* Adopted in 2012 (Cont.)

- Discourages nonmedical use of ultrasound for entertainment purposes
- Fetal heart rate should be analyzed with M-mode first before spectral Doppler is considered.

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Appropriate Use of ALARA

- Sonographers should:
  - Monitor output displays on equipment (thermal index)
  - Keep mechanical index (MI) and the TI as low as possible
  - Use TIS for soft tissue during 1st Tri.
  - Use TIB for bone after 10 weeks gestation (<0.7)
  - Move transducer frequently; do not dwell on one site

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Safety of Ultrasound

1. There are theoretical effects of ultrasound energy on fetus; there are potential biological effects that have not been documented.

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Safety of Ultrasound (Cont.)

2. The energy produced by ultrasound equipment today higher than that produced by earlier equipment. Doppler imaging produces higher energy. Not a long history of safety with energy at level in use today.

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Safety of Ultrasound (Cont.)

3. The studies of ultrasound bioeffects are not definitive; continued research essential
4. Sonographers have responsibility to be knowledgeable regarding ultrasound bioeffects and to use least amount of energy necessary to produce clinical information needed.

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Safety of Ultrasound (Cont.)

- AIUM adopted statement on *As Low As Reasonably Achievable (ALARA) principle* that states the following:
- The potential benefits and risks of each examination should be considered. The ALARA (As Low As Reasonably Achievable) Principle should be observed when adjusting controls that affect the acoustical output and by considering transducer dwell times.

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Safety of Doppler for the Obstetric Patient

- Doppler ultrasound provides noninvasive method to assess physiology and pathophysiology of fetal and maternal circulations when such examinations required for diagnosis
- In most cases, pulsed wave Doppler used in fetus rather than continuous wave Doppler

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Safety of Doppler for the Obstetric Patient (Cont.)

- Use of Doppler ultrasound during first trimester has generated some controversy because Doppler is higher energy level; these are difficult examinations that may require prolonged dwell times; and because embryo at this stage small and receives total body insonation

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Safety of Doppler for the Obstetric Patient (Cont.)

- Benefits of Doppler imaging most likely outweigh risks when there are specific indications requiring Doppler interrogation.
- Fetal sonography with or without Doppler should be performed only when there is valid medical reason, and lowest possible ultrasonic exposure settings should be used to obtain necessary diagnostic information.

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Guidelines for First Trimester and Standard Second and Third Trimester OB Sonography Examinations

- AIUM
- ACOG
- ACR
- Professional societies collaborate on clinical aspects of statements and include identical wording related to classification and specifications for examinations.
- Portions of statements including information related to physician qualification, procedure documentation, quality control, background, and clinical recommendations differ within organizations' statements.

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Guidelines for First Trimester and Standard Second and Third Trimester OB Sonography Examinations (Cont.)

- Guidelines often cited as legal standard, though this use was not intended
- While it is not possible to detect all fetal anomalies with sonography, adherence to practice guidelines and referral of any suspicious studies for further evaluations will optimize possibilities of detection.

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Guidelines for First Trimester and Standard Second and Third Trimester OB Sonography Examinations (Cont.)

- Important that all sonographers strive to consistently meet or exceed minimum standards during every obstetric sonographic examination.

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Guidelines for First Trimester and Standard Second and Third Trimester OB Sonography Examinations (Cont.)

- Quality standards
  - Components of examination protocol
  - Qualifications of personnel performing examination
  - Documentation
  - Equipment specifications
  - Fetal safety
  - Quality control
  - Safety
  - Infection control
  - Patient education concerns

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Guidelines for First Trimester and Standard Second and Third Trimester OB Sonography Examinations (Cont.)

- National board certification—RDMS
- Purpose of certification or registry is to assure public that person performing sonography has necessary knowledge, skills, and experience to provide this service.

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Guidelines for First Trimester and Standard Second and Third Trimester OB Sonography Examinations (Cont.)

- Documentation *guidelines recommend* that there is permanent record maintained of measurements and anatomic findings. Images need to be labeled with patient's name, date, and image orientation if appropriate.

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Guidelines for First Trimester and Standard Second and Third Trimester OB Sonography Examinations (Cont.)

- Written report of examination must be maintained in patient records.
- Availability of real-time sonography equipment with transabdominal and transvaginal transducers essential to confirm fetal life and to permit sonographer to view fetal anatomy and movements and to obtain biometric parameters used to determine fetal age and growth.

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Guidelines for First Trimester and Standard Second and Third Trimester OB Sonography Examinations (Cont.)

- Transducer frequency selection and equipment settings should balance optimal imaging resolution and penetration.
- Lowest possible exposure setting should be used according to ALARA (As Low As Reasonably Achievable) principle.

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Guidelines for First Trimester and Standard Second and Third Trimester OB Sonography Examinations (Cont.)

- Policies and procedures related to patient information, infection control, quality control, and safety should be developed and implemented in every laboratory.
- Policies typically address personnel and patient safety and may include policies that address musculoskeletal injury concerns for sonographer.

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First Trimester

- Either transabdominal or transvaginal transducer may be used for examination of first trimester embryo and fetus.
- If transabdominal examination not definitive, transvaginal or transperineal examination required.

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First Trimester (Cont.)

- Transabdominal examination may provide overview of entire pelvic cavity.
- Enable sonographer to image
  - Uterus from cervix to fundus
  - Evaluate ovaries
  - Adnexal areas for abnormal collections of fluid or mass
  - Look for presence of free fluid

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First Trimester (Cont.)

- Transvaginal transducer provides more limited view of pelvic cavity.
- Allows excellent visualization of:
  - Embryo
  - Yolk sac
  - Amnion
  - Chorion
  - Gestational sac

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First Trimester Sample Protocol

1. Uterus and adnexa should be evaluated for presence of gestational sac.

- If gestational sac seen, location (intrauterine or extrauterine) should be noted.
- Gestational sac should be recorded when embryo not identified during zygote or implantation stage of pregnancy.
- Caution must be used in diagnosing gestational sac without presence of yolk sac or embryo since intrauterine fluid collections may appear similar.

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First Trimester
Sample Protocol (Cont.)

- Presence or absence of yolk sac and embryo should be noted.
- Crown-rump length most accurate measurement of gestational age during first trimester
  - Should be recorded when embryo present

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First Trimester Sample Protocol (Cont.)

- Earliest structure seen within gestational sac is yolk sac; yolk sac will indicate presence of intrauterine pregnancy.
- Embryo seen at 4 weeks as echogenic curved structure adjacent to yolk sac

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First Trimester
Sample Protocol (Cont.)

- Blood tests (hCG levels) should be positive at 7 to 10 days embryonic age (conception age).
- Placenta seen as thickened density (trophoblastic reaction) along part of margin of gestational sac

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First Trimester
Sample Protocol (Cont.)

1. Bowel herniates out from fetal abdomen at 8 to 11 weeks, then returns to abdominal cavity.

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First Trimester
Sample Protocol (Cont.)

2. Presence or absence of cardiac activity should be reported.

- Cardiac motion usually seen when embryo 5 mm or greater in length.
- Fetal heart rate much faster than mother's heart rate
  - Fetal HR changes according to fetal development stages; early in embryologic development HR slow (90 bpm)
  - Rate may go up to 180 bpm in middle of first trimester, before returning to 120 to 160 bpm throughout remainder of pregnancy.

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First Trimester
Sample Protocol (Cont.)

3. Fetal number should be documented.

- Count only embryo and yolk sac to determine multiple pregnancies. Membrane structure and number of amniotic and chorionic membranes should be documented in all multiple pregnancies. Chorionicity most reliably documented during first trimester

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First Trimester
Sample Protocol (Cont.)

4. Evaluation of uterus, adnexal structures, and cul-de-sac should be performed.

- Important to document texture of ovaries and presence of corpus luteum or other adnexal masses; look for inhomogeneous uterine texture that may represent leiomyomatous growth stimulated by hormonal changes of pregnancy

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First Trimester
Sample Protocol (Cont.)

5. During late first trimester (11 to 14 weeks), a nuchal translucency measurement may be performed according to standard criteria for women desiring to assess their individual risk for fetal aneuploidy.

- Fetal anatomy survey
- *This fetal anatomy survey cannot replace the second trimester survey.*

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Indications for Second and Third Trimester Sonography

- Evaluation of gestational age
- Evaluation of fetal growth
- Evaluation of vaginal bleeding
- Evaluation of cervical insufficiency
- Evaluation of abdominal and pelvic pain
- Determination of fetal presentation
- Evaluation of suspected multiple gestation

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Indications for Second and Third Trimester Sonography (Cont.)

- Adjunct to amniocentesis or other procedure
- Significant discrepancy between uterine size and clinical dates
- Evaluation of pelvic mass
- Examination of suspected hydatidiform mole
- Adjunct to cervical cerclage placement
- Evaluation of suspected ectopic pregnancy

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Indications for Second and Third Trimester Sonography (Cont.)

- Evaluation of suspected fetal death
- Evaluation of suspected uterine abnormality
- Evaluation of fetal well-being
- Evaluation of suspected amniotic fluid abnormalities
- Evaluation of suspected placental abruption

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Indications for Second and Third Trimester Sonography (Cont.)

- Adjunct to external cephalic version
- Evaluation for premature rupture of membranes and/or premature labor
- Evaluation for abnormal biochemical markers
- Follow-up evaluation of fetal anomaly

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Indications for Second and Third Trimester Sonography (Cont.)

- Follow-up evaluation of placental location for suspected placenta previa
- Evaluation of those with history of previous congenital anomaly
- Evaluation of fetal condition in late registrants for prenatal care
- To assess findings that may increase risk of aneuploidy
- To screen for fetal anomalies

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Protocol for Second and Third Trimester Sonography

1. Fetal cardiac motion, fetal number, presentation, activity should be documented.

- In multiple gestations, the following individual studies should be performed on each fetus: amnionicity, chorionicity, comparison of fetal sizes, estimation of amniotic fluid (increased, decreased, or normal) on each side of membrane, fetal gender (when visualized)

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Protocol for Second and Third Trimester Sonography (Cont.)

2. Qualitative or semiquantitative estimate of volume of amniotic fluid should be reported. Abnormal fluid amounts should be described.

- In early pregnancy amniotic fluid produced by placenta; fetal kidneys begin to produce urine, which contributes to production and replacement of amniotic fluid as fetus swallows and urinates. Fluid increases in volume until 34th week of gestation.

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Protocol for Second and Third Trimester Sonography (Cont.)

- Experienced observers may subjectively estimate amniotic fluid volume. Semiquantitative methods, including four-quadrant amniotic fluid index (AFI), single deepest pocket, and two-diameter pocket may be used.
- Excessive fluid termed hydramnios (polyhydramnios); too little fluid called oligohydramnios.

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Protocol for Second and Third Trimester Sonography (Cont.)

3. Placental localization, appearance, relationship to internal cervical os should be recorded. Umbilical cord should be imaged and number of vessels should be evaluated when possible.

- Maternal bladder must be adequately filled to see cervical os in lower uterine segment. Should document that lower end of placenta is away from cervical os to rule out placenta previa.

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Protocol for Second and Third Trimester Sonography (Cont.)

- Overdistended maternal urinary bladder or contraction in lower uterine cavity can give false impression of placenta previa.
- Placental location in early pregnancy may not correlate well with its location at time of delivery. Transvaginal imaging may be helpful in visualizing the internal os and the position of the placenta in relation to the cervix.

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Protocol for Second and Third Trimester Sonography (Cont.)

- Transvaginal or transperineal imaging may be necessary to document cervical length if there are risk factors for spontaneous preterm birth when the cervix appears shortened or if history of regular uterine contractions exists.
- Value of routine cervical length measurements in low risk pregnancy has not been established.

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Protocol for Second and Third Trimester Sonography (Cont.)

4. Gestational (menstrual) age should be assessed by sonographic biometry. CRL measured during first trimester is most accurate method to assess gestational age. During second and third trimesters, there are multiple sonographic parameters that can be used to estimate gestational age.

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Protocol for Second and Third Trimester Sonography (Cont.)

Variability of measurements increases as pregnancy progresses. If clinical gestational age and sonographic parameters demonstrate significant discrepancies, possibility of fetal growth abnormalities, such as macrosomia or IUGR, suggested. Biometric parameters that may be used include following:

- Biparietal diameter (BPD) measured in axial plane that includes thalamus and cavum septum pellucidum

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Protocol for Second and Third Trimester Sonography (Cont.)

- Cerebellar hemispheres should not be visible in image where BPD measured.
- Biparietal measurement taken from outer edge to inner edge of skull
- Head shape may normally be more rounded (brachycephaly) or elongated (dolichocephaly), which makes measurement of head circumference more accurate than measurement of BPD.

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Protocol for Second and Third Trimester Sonography (Cont.)

- Head circumference measured at same level as BPD, around outer perimeter of calvarium. HC not affected by head shape
- Femur length and femoral diaphysis length are reliably measured after 14th week of gestation. Most accurate measurements taken when femoral shaft perpendicular to acoustic beam

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Protocol for Second and Third Trimester Sonography (Cont.)

- Abdominal circumference measured on transverse view at level of junction of umbilical vein and portal sinus
- Circumference should be measured at skin line on true transverse view of fetal abdomen where portal sinus, fetal stomach, and umbilical vein visible; abdominal circumference used to estimate fetal weight

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Protocol for Second and Third Trimester Sonography (Cont.)

- Fetal weight may be estimated by using AC measurement in mathematical formulations with other sonographic parameters.
- Best fetal weight estimates yield significant errors. Interval growth may be determined by sonographic measurements 2 to 4 weeks apart.

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Protocol for Second and Third Trimester Sonography (Cont.)

5. Evaluation of maternal anatomy including uterine, adnexal, cervical evaluation should be performed to document presence, location, and size of uterine or adnexal masses, which may complicate obstetric management. Normal maternal ovaries may not be imaged during 2nd and 3rd trimester.

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Protocol for Second and Third Trimester Sonography (Cont.)

6. Fetal anatomy may be adequately assessed after 18 weeks of gestation. Anatomy may be difficult to image because of fetal movement, size, or position; maternal scars; or increased wall thickness. When anatomy not seen because of technical limitation, sonographer should note reason. A follow-up examination may be ordered.

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Protocol for Second and Third Trimester Sonography (Cont.)

- Practice Guidelines recommend that the following anatomy be documented during standard obstetric sonography examination.
- More detailed studies may be performed if anatomy appears questionable or abnormal. Documentation and images of required anatomy should be retained.

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Protocol for Second and Third Trimester Sonography (Cont.)

1. Head and neck
   - Cerebellum
   - Choroid plexus
   - Cisterna magna
   - Lateral cerebral ventricles
   - Midline falx
   - Cavum septi pellucidi
   - Upper lip
   - Measurement of the nuchal fold

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Protocol for Second and Third Trimester Sonography (Cont.)

2. Chest
   - Four-chamber view of fetal heart
   - Left ventricular outflow tract
   - Right ventricular outflow tract

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Protocol for Second and Third Trimester Sonography (Cont.)

3. Abdomen
   - Stomach (presence, size, situs)
   - Kidneys
   - Urinary bladder
   - Umbilical cord insertion into fetal abdomen
   - Umbilical cord; number of vessels

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Protocol for Second and Third Trimester Sonography (Cont.)

4. Spine
   - Cervical, thoracic, lumbar, sacral spine
5. Extremities
   - Presence or absence of arms and legs
6. Gender
   - In multiple pregnancies and when medically indicated

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Diagnostic and Screening Aspects of Obstetric Sonography

- Diagnostic tests can give definitive information about clinical question or presence or absence of finding.
- Obstetric sonography examinations typically diagnostic with respect to fetal heart motion, fetal number, fetal biometry, fetal presentation, location of placenta, presence of maternal pelvic or adnexal mass, major disruptions of fetal anatomy such as anencephaly.

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Diagnostic and Screening Aspects of Obstetric Sonography (Cont.)

- Screening test does not provide definitive diagnosis but indicates if patient or pregnancy at greater or lesser risk.
- Obstetric sonography examination with normal findings may reduce risk of that fetus having anomaly but does provide certainty that fetus not affected.

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Diagnostic and Screening Aspects of Obstetric Sonography (Cont.)

- Sensitivity depends on many factors including maternal habitus, expertise of sonographers and physicians responsible for study, risk level of patients, number and timing of sonography examinations, and type of anomaly.

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Diagnostic and Screening Aspects of Obstetric Sonography (Cont.)

- Sensitivity tends to be higher for defects of central nervous system and urinary system and lower for defects of heart and great vessels.
- Sensitivity higher in tertiary care centers and with specialty obstetric examinations
- Maternal obesity known to impact sonographic visualization of anatomy

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Diagnostic and Screening Aspects of Obstetric Sonography (Cont.)

- Important that patients be counseled about limitations of obstetric sonography
- Counseling should include information that screening examinations do not detect all anomalies, that false-positive findings possible, and that sensitivity of sonography is not certain in any situation.

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